

실습 #7

Lab7_1

Lab7_1.java 코드

```
import java.awt.*;
import javax.swing.*;

public class Lab7_1{
    public static void main(String[] args) {
        int numCircles = 3;

        if (args.length > 0) {
            numCircles = Integer.parseInt(args[0]);
            if (numCircles < 3) {
                numCircles = 3;
            }
        }

        JFrame frame = new JFrame("Lab7_1");
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.getContentPane().setLayout(new BorderLayout());

        JPanel answer = new JPanel();
        answer.setBackground(Color.yellow);
        answer.setPreferredSize(new Dimension(250, 75));

        JButton yes = new JButton("YES");
        JButton no = new JButton("NO");
        JLabel label = new JLabel("Choose YES or NO");

        answer.add(yes);
        answer.add(no);
        answer.add(label);

        TrafficLightPanel traffic = new TrafficLightPanel(numCi
```

```

rcles);

    frame.getContentPane().add(answer, BorderLayout.NORTH);
    frame.getContentPane().add(traffic, BorderLayout.CENTE
R);

    frame.pack();
    frame.setVisible(true);
}
}

class TrafficLightPanel extends JPanel {
    private int numCircles;

    public TrafficLightPanel(int numCircles) {
        this.numCircles = numCircles;
        setPreferredSize(new Dimension(50 * numCircles + 50, 20
0));
        setBackground(Color.black);
    }

    public void paintComponent(Graphics page) {
        super.paintComponent(page);

        int x = 50;

        for (int i = 0; i < numCircles; i++) {
            if (i == 0) {
                page.setColor(Color.red);
            } else if (i == 1) {
                page.setColor(Color.yellow);
            } else if (i == 2) {
                page.setColor(Color.green);
            } else {
                page.setColor(Color.blue);
            }

            page.fillOval(x, 50, 40, 40);

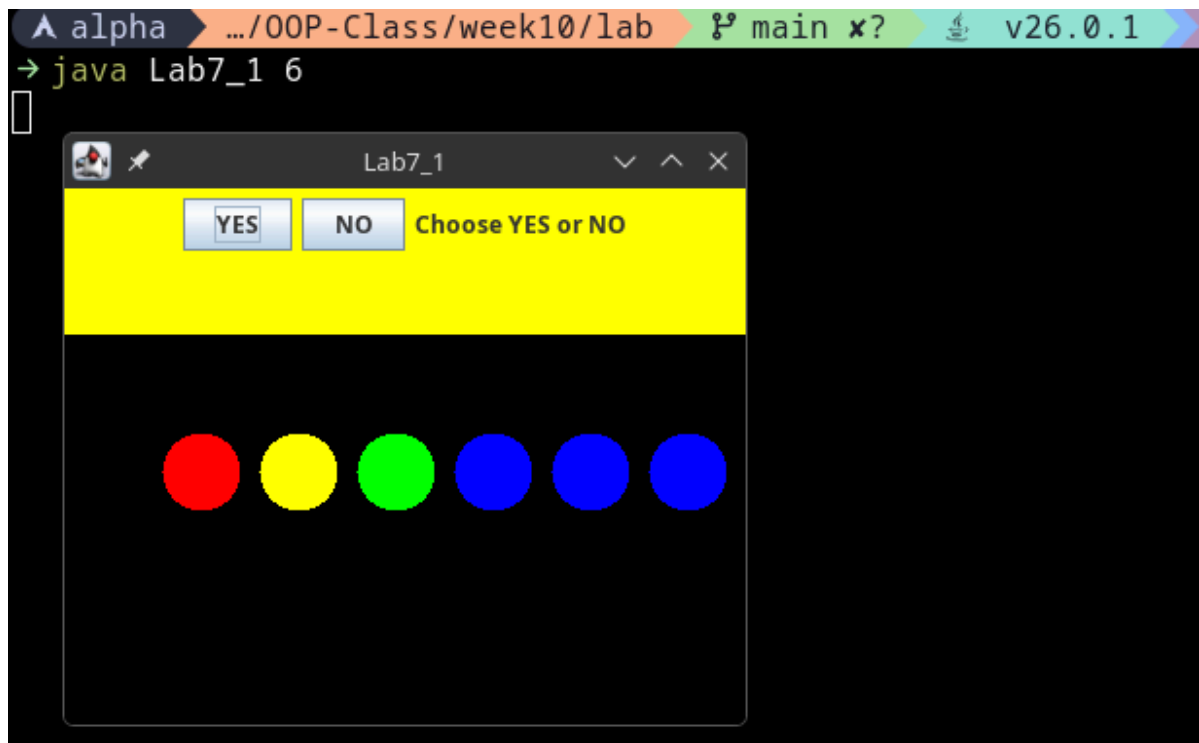
```

```

        x += 50;
    }
}
}

```

실행 화면



Lab7_2

Lab7_2.java 코드

```

import java.awt.*;
import javax.swing.*;

public class Lab7_2 {
    public static void main (String[] args) {
        JFrame frame = new JFrame ("Lab7_2");
        frame.setDefaultCloseOperation (JFrame.EXIT_ON_CLOSE);
        frame.getContentPane().add(new MyPanel());
        frame.pack();
        frame.setVisible(true);
    }
}

```

```

    }
}

class MyPanel extends JPanel {
    public MyPanel() {
        setPreferredSize (new Dimension(300, 200));
        setBackground (Color.cyan);

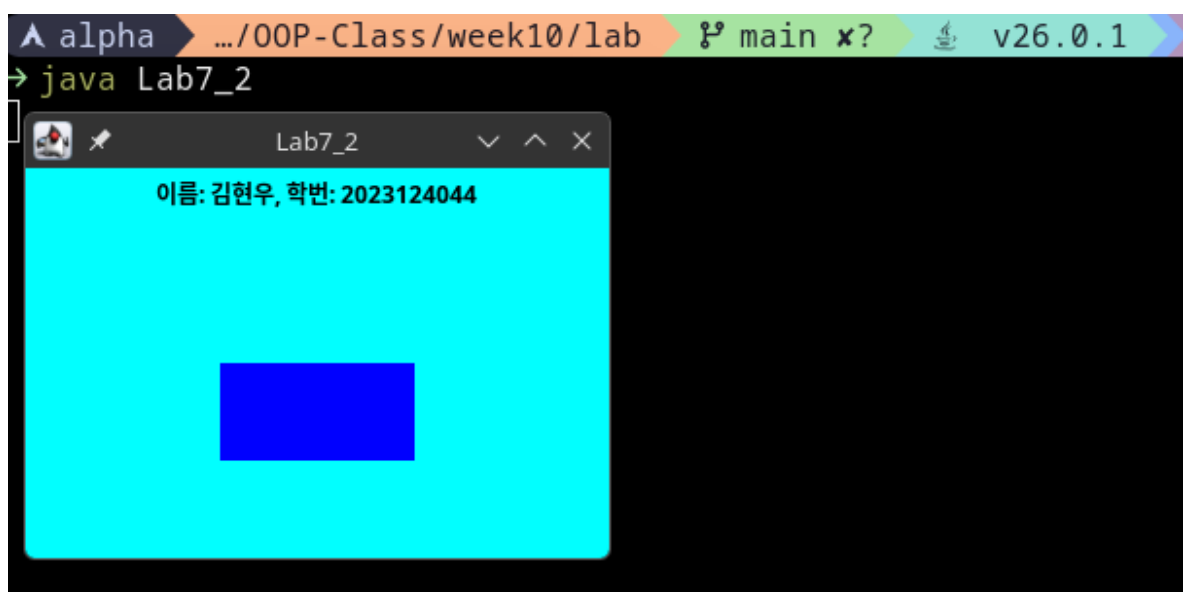
        JLabel label = new JLabel("이름: 김현우, 학번: 202312404
4");
        label.setForeground(Color.black);
        add(label);
    }

    public void paintComponent(Graphics page) {
        super.paintComponent(page);

        page.setColor(Color.blue);
        page.fillRect(100, 100, 100, 50);
    }
}

```

실행 화면



Lab7_3

Lab7_3.java 코드

```
import java.awt.Color;
import java.awt.Container;
import java.awt.Dimension;
import java.awt.Graphics;
import java.util.ArrayList;
import javax.swing.JFrame;
import javax.swing.JPanel;

public class Lab7_3 {
    public static void main(String [] args) {
        JFrame frame = new JFrame ("Lab7_3");
        frame.setDefaultCloseOperation (JFrame.EXIT_ON_CLOSE);
        Container cp = frame.getContentPane();
        cp.add (new RubberLinePanel());
        frame.pack();
        frame.setVisible(true);
    }
}

class RubberLinePanel extends JPanel
{
    ArrayList<Polygon> plist = new ArrayList<Polygon>();
    RubberLinePanel() {
        Circle c1= new Circle(10,10,100);
        Circle c2 = new Circle(20,20,50);
        Rectangle r1 = new Rectangle(20,20,100,200);
        Rectangle r2 = new Rectangle(100,100,300,50);
        plist.add(c1);
        plist.add(c2);
        plist.add(r1);
        plist.add(r2);

        setPreferredSize (new Dimension(1000, 1000));
        add (new DrawPanel());
    }
}
```

```

}
private class DrawPanel extends JPanel {
    public DrawPanel(){
        setBackground (Color.black);
        setPreferredSize (new Dimension(500, 500));
    }

    public void paintComponent(Graphics page) {
        super.paintComponent(page);
        page.setColor(Color.yellow);

        Polygon p;
        for(int i = 0; i < plist.size(); i++) {
            p = plist.get(i);
            if (p instanceof Circle) {
                Circle c = (Circle) p;
                page.drawOval(c.getX(), c.getY(), c.getRadius() *
2, c.getRadius() * 2);
            } else if (p instanceof Rectangle) {
                Rectangle r = (Rectangle) p;
                page.drawRect(r.getX(), r.getY(), r.getWidth(),
r.getHeight());
            }
        }
    }
}

```

(Polygon.java는 6주차 7주차 과제의 파일을 사용함)

실행 화면

